

Lecture Notes

Stochastic Hydrology

Module 2: Frequency Analysis

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Preface

These lecture notes are intended for personal use in the Stochastic Hydrology course and may not be distributed further.

4 Frequency Analysis

4.1 Definitions

Frequency analysis is applicable to a wide range of problems, including runoff volumes, low flows, rainfall events, and water quality parameters.

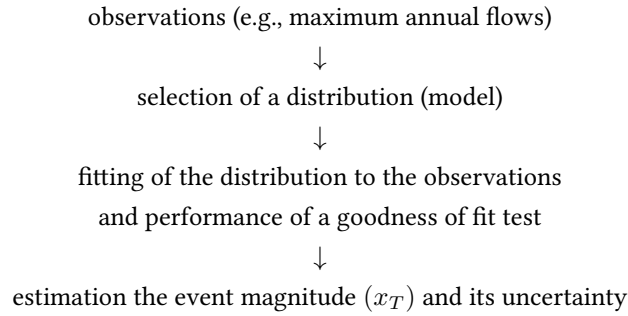
There are two assumptions about the data that have to hold for the performance of frequency analysis:

- *Homogeneity*: All observations come from the same population.
- *Independence*: There is no correlation between individual observations, each event is random and separate.

Frequency analysis is used for two scenarios:

- Estimate the magnitude of an event with a given frequency of occurrence.
- Estimation of the frequency of occurrence of a given event.

Procedure for frequency analysis:



Flood frequency analysis is usually performed on the following data:

- *Maximum annual series*: The one maximum observation from each year is selected.
- *Partial duration series*: All observations above a threshold are selected.

4.1.1 Return Period

The *return period* T describes the average time (in years) between the occurrences of an event of a specific magnitude x_T . The return period is related to the probability p of an event occurring in any given year.

$$T = \frac{1}{p} \quad (4.1)$$

$$\mathbb{P}(X \geq x_T) = \frac{1}{T} = p \quad \text{probability of exceedance} \quad (4.2)$$

$$\mathbb{P}(X < x_T) = 1 - \frac{1}{T} = 1 - p \quad \text{cumulative probability / prob. of non-exceedance} \quad (4.3)$$

4.2 Distributions in Frequency Analysis

There are several *extreme value distributions* commonly used to describe maximum value data, including the Gumbel / EV-I distribution, Generalized Extreme Value Distribution GEV, and Log Pearson Type III distribution.

4.2.1 Gumbel Distribution / EV-I

Given N random variables $Y_1, Y_2, \dots, Y_N$ (e.g., set of daily streamflow values), we define the maximum over a given period as $X = \max\{Y_1, Y_2, \dots, Y_N\}$ (e.g., maximum annual streamflow value).

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(\text{all } Y\text{'s} \leq x) \quad (4.4)$$

If we assume that the random variables Y_i are independent:

$$\begin{aligned} F_X(x) &= \mathbb{P}(Y_1 \leq x) \cdot \mathbb{P}(Y_2 \leq x) \cdot \dots \cdot \mathbb{P}(Y_N \leq x) \\ &= F_{Y_1}(x) \cdot F_{Y_2}(x) \cdot \dots \cdot F_{Y_N}(x) \end{aligned} \quad (4.5)$$

If the random variables Y_i are further more identically distributed:

$$F_X(x) = [F_Y(x)]^N \quad (4.6)$$

If Y_i is a continuous random variable:

$$f_X(x) = \frac{d}{dx} F_X(x) \quad (4.7)$$

$$= N \cdot [F_Y(x)]^{N-1} \cdot f_Y(x) \quad (4.8)$$

The distribution of the extreme value depends on N and on the parent distribution. As $N \rightarrow \infty$, limiting forms of $f_X(x)$ can be found. The Gumbel distribution / EV-I arises as the distribution of the largest of N values of Y_i as $N \rightarrow \infty$. The parent distribution $f_Y(y)$ is unbounded $y > 0$ and the upper tail falls off in an exponential manner. In the tail, we can assume:

$$F_Y(y) = 1 - e^{-g(y)} \quad (4.9)$$

where $g(y)$ describes the increasing form of y (e.g., if $g(y) = \lambda y$, exponential distribution). Performing a Taylor series expansion and approaching the limit $N \rightarrow \infty$:

$$F_X(x) = e^{-e^{-\alpha(x-\beta)}} = \exp\{-\exp[-\alpha(x-\beta)]\} \quad -\infty < x < \infty \quad (4.10)$$

The scale parameter α and location parameter β are estimated from the data.

Method of Moments For the Gumbel / EV-I distribution, the method of moments offers the easiest way to obtain the parameters.

$$\mu_X = \beta + \frac{0.5772}{\alpha} \quad (4.11)$$

$$\sigma_X^2 = \frac{\pi^2}{6 \cdot \alpha^2} \quad (4.12)$$

$$\gamma_X = 1.1396 \quad (4.13)$$

$$K_X = 5.40 \quad (4.14)$$

Solving for the Gumbel / EV-I parameters results in

$$\hat{\alpha} = \sqrt{\frac{\pi^2}{6 \cdot S_X^2}} = \frac{1.2825}{S_X} \quad (4.15)$$

$$\hat{\beta} = \bar{x} - \frac{0.5772}{\hat{\alpha}} = \bar{x} - 0.5772 \cdot \frac{S_X}{1.2825} = \bar{x} - 0.45 \cdot S_X \quad (4.16)$$

Maximum Likelihood Method Solve for $\hat{\alpha}$ from

$$F(\hat{\alpha}) = \sum_{i=1}^N x_i \cdot e^{-\hat{\alpha} \cdot x_i} - \left(\bar{x} - \frac{1}{\hat{\alpha}} \right) \cdot \sum_{i=1}^N e^{-\hat{\alpha} \cdot x_i} = 0$$

$$\hat{\beta} = \frac{1}{\hat{\alpha}} \cdot \ln \left(\frac{N}{\sum_{i=1}^N e^{-\hat{\alpha} \cdot x_i}} \right)$$

Goodness of Fit To obtain the *goodness of fit* for the Gumbel / EV-I probability plot, first order the observations from smallest to largest. The plotting position of the i^{th} observation is described by the Gringorton plotting position q_i :

$$q_i = \frac{i - 0.44}{N + 0.12} \quad (4.17)$$

Plot the observations x_i against their expected value $x_i(q_i)$:

$$q_i = F_X(x_i) = e^{-e^{-\alpha(x_i - \beta)}} \quad (4.18)$$

$$\rightarrow x_i(q_i) = \hat{\beta} - \frac{1}{\hat{\alpha}} \ln[-\ln(q_i)] \quad (4.19)$$

The observed and expected data values can be used to additionally perform a PPCC test.

Estimation of T -Year Event The *magnitude* x_T of an event with a return period of T years under the Gumbel / EV-I distribution can be estimated as follows:

$$\mathbb{P}(X < x_T) = 1 - \frac{1}{T} \quad (4.20)$$

$$F_X(x_T) = 1 - \frac{1}{T} = e^{-e^{-\hat{\alpha}(x_T - \hat{\beta})}} \quad (4.21)$$

$$\rightarrow x_T = \hat{\beta} - \frac{1}{\hat{\alpha}} \ln \left[-\ln \left(1 - \frac{1}{T} \right) \right] \quad (4.22)$$

Confidence Interval Based on a given sample of data, we are able to estimate the corresponding event magnitude x_T for a return period of T years. If another sample were to be available, we would obtain different estimates for the parameters and thus a different x_T estimate. If we had many samples, we could get an idea of the range of values x_T can take on. Since we often only have one sample, statistical procedures can be employed to estimate the confidence interval about the estimate x_T .

The *confidence interval around the mean* describes the range where the true mean will be contained between a lower l and upper u value $(1 - \alpha)\%$ of the time. $(1 - \alpha)$ describes the confidence level, which is the probability that the interval includes the true mean.

$$\mathbb{P}(l \leq \mu \leq u) = 1 - \alpha \quad (4.23)$$

We can find l and u as a function of the sample mean $\bar{x}$, the sample standard deviation S_X and the significance level α :

$$\frac{\bar{x} - \mu}{\sqrt{S_x^2/n}} \sim t \quad t\text{-distribution with } (n-1) \text{ degrees of freedom}$$

$$\frac{\bar{x} - \mu}{\sqrt{S_x^2/n}} \sim \mathcal{N}(0, 1) \quad \text{for a large } n (>30), \text{ standard normal distribution}$$

$$\mathbb{P} \left(-t_{1-\alpha/2, n-1} \leq \frac{\bar{x} - \mu}{\sqrt{S_X^2/n}} \leq t_{1-\alpha/2, n-1} \right) = 1 - \alpha \quad (4.24)$$

The interval around the mean μ can be found by rearranging the previous equation:

$$\mathbb{P} \left(\bar{x} - t_{1-\alpha/2, n-1} \cdot \sqrt{\frac{S_X^2}{n}} \leq \mu \leq \bar{x} + t_{1-\alpha/2, n-1} \cdot \sqrt{\frac{S_X^2}{n}} \right) = 1 - \alpha \quad (4.25)$$

The table of the standard normal distribution z and the t -distribution are given in Tables A.1 and A.2 in the Appendix.

The *confidence interval on design events* describes the range for which the true event magnitude will be contained between the lower x_T^l and upper x_T^u value $(1 - \alpha)\%$ of the time.

If we assume that the probability density function of x_T (not that of x) is normal, then

$$x_T^{u, l} = x_T \pm S(x_T) \cdot z_{1-\alpha/2} \quad (4.26)$$

where $S(x_T)$ describes the standard error of the estimate x_T . The standard error depends on the distribution of x , the method of fitting, the number of observations, and the return period (see Fig. 4.1).

- Gumbel / EV-I distribution & Method of Moments:

$$S(x_T) = \frac{\delta}{\sqrt{n}} \cdot S_X \quad (4.27)$$

where the parameter δ is provided in Table A.4 for a set of return periods T and sample sizes n . Alternatively, δ can be approximated as

$$\delta = \sqrt{1 + 1.1396K_T + 1.1K_T^2} \quad \text{with } K_T = \frac{y_T - \mu_y}{\sigma_y} \quad (4.28)$$

with the reduced variate $y_T = -\ln[-\ln(1 - 1/T)]$ and values of μ_n and σ_n as a function of sample size n provided in Table A.3 in the Appendix.

- Gumbel / EV-I distribution & Method of Maximum Likelihood:

The standard error $S(x_T)$ is a function of the parameter δ_m , which is provided in Table A.5 in the Appendix for different return periods T .

$$S(x_T) = \delta_m \sqrt{\frac{S_X^2}{n}} \quad (4.29)$$

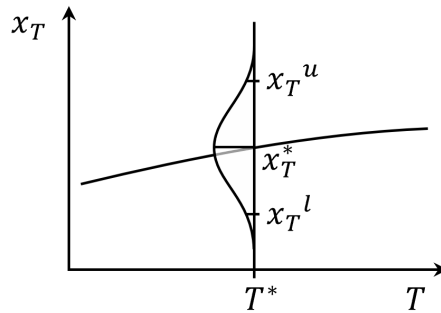


Figure 4.1. Visualization of the confidence interval on the design event x_T^* with return period T^* . The probability of the true event magnitude falling into the confidence interval $[x_T^l, x_T^u]$ is $1 - \alpha$.

4.2.2 Generalized Extreme Value Distribution GEV

Distributions of the extreme values selected from sets of samples of any probability distribution have been shown to converge to one of three forms of extreme value distributions, called Types I, II and III, respectively, when the number of selected extreme values is large. The properties of the three limiting forms were further developed by Gumbel for the Extreme Value Type I (EV-I) distribution, Fréchet for the Extreme Value Type II (EV-II), and Weibull for the Extreme Value Type III (EV-III). These three forms were also shown to be special cases of the *Generalized Extreme Value distribution GEV*. Its probability and cumulative distribution function are given in Equations 4.30 and 4.31. Note that the parameters $-\infty < \beta < \infty$, $\alpha > 0$ and $-\infty < \kappa < \infty$ refer to the location, scale, and shape parameters, respectively.

$$f_X(x) = \frac{1}{\sigma} [1 + \kappa \alpha (x - \beta)]^{-\frac{1}{\kappa} - 1} \cdot \exp \left\{ - [1 + \xi \alpha (x - \beta)]^{-\frac{1}{\kappa}} \right\} \quad \text{for } \xi \neq 0 \quad (4.30)$$

$$F_X(x) = \exp \left\{ - [1 - \xi \alpha (x - \beta)]^{1/\xi} \right\} \quad \text{for } \xi \neq 0 \quad (4.31)$$

The three types are differentiated by the shape parameter κ :

- Gumbel distribution / EV-I: $\kappa = 0$
- Fréchet distribution / EV-II: $\kappa > 0$, supported for $x > \beta - 1/(\alpha \kappa)$
- Weibull distribution / EV-III: $\kappa < 0$, supported for $x < \beta - 1/(\alpha \kappa)$

You can find information on the parameter estimation for the GEV distribution for example in the textbook *Statistical Methods in the Atmospheric Sciences* (Wilks 2011).

5 Multivariate Frequency Analysis

5.1 Definitions

Compound hazards refer to situations where multiple factors interact to produce significant impacts. These can involve two or more simultaneous or successive extreme events, such as consecutive storms or a heatwave followed by drought. They may also arise from combinations of extreme events with underlying conditions—like high population vulnerability or degraded infrastructure—that amplify the overall impact. In addition, compound hazards can result from combinations of events that are not individually extreme but, when occurring together, lead to an extreme event or severe consequences.

Classification of compound events:

- *Preconditioned*: One or more hazards can cause an impact, or lead to an amplifying impact, only because of a preexisting, climate-driven condition (e.g., heavy precipitation or saturated soil).
- *Multivariate*: Co-occurrence of multiple climate drives and / or hazards in the same geographical region causing an impact.
- *Temporally compounding*: Succession of hazards that affect a given region leading to or amplifying an impact when compared to a single hazard (e.g., succession of heatwaves).
- *Spatially compounding*: Multiple connected locations are affected by the same or different hazards within a limited time window (e.g., spatially compounding precipitation events).

Flood mechanisms:

- *Pluvial*: Flooding caused by intense rainfall overwhelming drainage systems, leading to water accumulation on the surface, often in urban areas.
- *Fluvial*: Flooding that occurs when river or reservoir storage exceeds their banks / levees / dams due to excessive rainfall or upstream inflow.
- *Coastal*: Flooding in coastal areas or oceans, seas of lakes, driven processes such as storm surges, high tides, and sea level rise.

5.1.1 Probability of Compound Hazards

- $\mathbb{P}(A \cap B)$: Probability of event A intersected with event B.
- $\mathbb{P}(B | A)$: Probability of an event B occurring given that A has occurred.

$$\mathbb{P}(A \cap B) = \mathbb{P}(A | B) \cdot \mathbb{P}(B) = \mathbb{P}(B | A) \cdot \mathbb{P}(A)$$

$$\begin{aligned} \text{If } A \text{ and } B \text{ are independent:} \quad & \mathbb{P}(A | B) = \mathbb{P}(A) \\ & \mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B) \end{aligned}$$

In the case of compound hazards it is possible that events A and B are not independent, and therefore $\mathbb{P}(A \cap B) \neq \mathbb{P}(A) \cdot \mathbb{P}(B)$. In this case, we need multivariate statistical models to characterize the risk associated with compound events. These can generalize univariate distributions to *joint distributions* which provide information about the relative likelihood that a set of multiple random variables will take on particular values. We refer to the univariate expression for the distribution of a single random variable as a *marginal distribution* which provides the distribution of a random variable irrespective of the values of the other random variables.

Joint probability density functions PDF and cumulative distribution functions CDF are expressed below. Figure 5.1 visualizes a bivariate distribution.

$$f_{XY}(x, y) = \mathbb{P}(x < X \leq x + dx \cap y < Y \leq y + dy) \quad (5.1)$$

$$F_{XY}(x, y) = \mathbb{P}(X \leq x \cap Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{XY}(\alpha, \beta) d\alpha d\beta \quad (5.2)$$

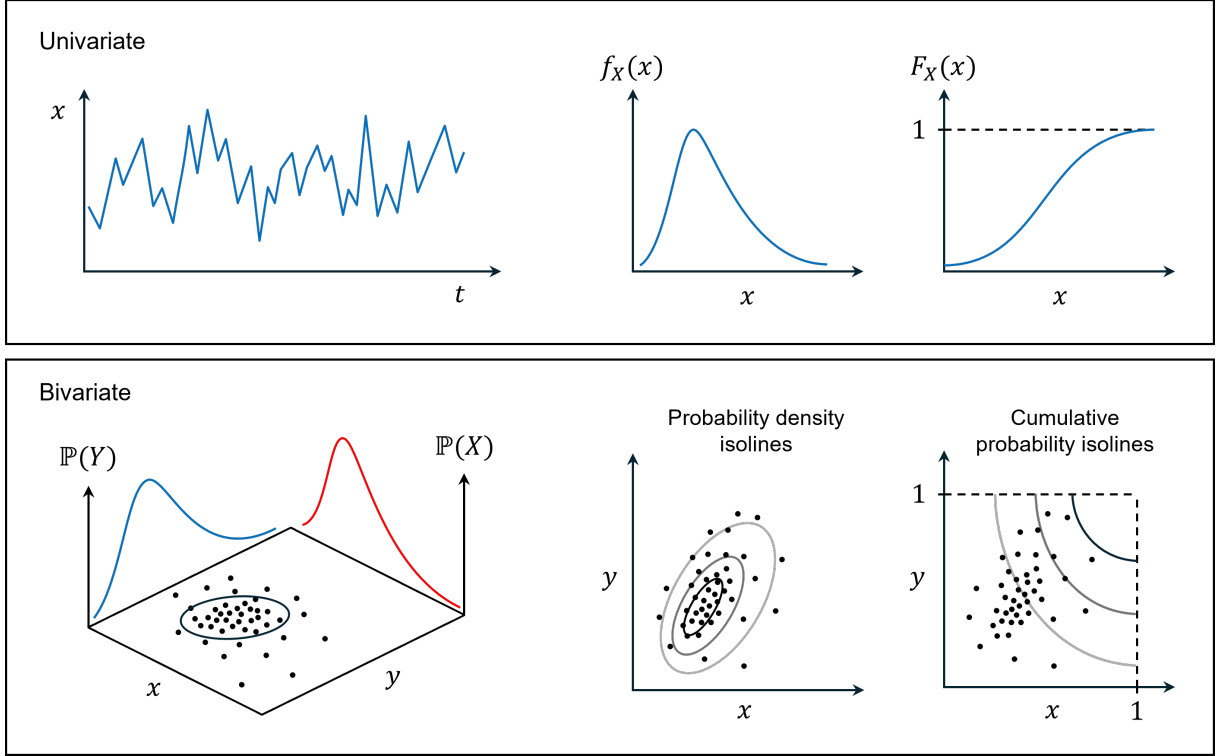


Figure 5.1. Visualization of a univariate (top) and bivariate (bottom) distribution. For the univariate distribution, we show a timeseries (left), probability density function $f_X(x)$ (middle) and cumulative probability distribution $F_X(x)$ (right). For the bivariate distribution, the joint behaviour of X and Y in relation to their marginal distributions (left), the probability density isolines (middle) and the cumulative probability isolines (right) are depicted.

The *marginal* PDF and CDF for the random variable X are described below and similarly hold for Y . They are visualized in Figure 5.1.

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy \quad (5.3)$$

$$F_X(x) = \int_{-\infty}^x f_X(\beta) d\beta \quad (5.4)$$

5.2 Approaches to Quantify Multi-Hazard Risk

There are three groups of approaches to quantify multi-hazard risk:

- *Structure Variable Approach* (continuous simulations): A multivariate input data is transformed into equivalent structure variable values which are then analyzed.

Example: Generate thousands of simulations of a flood level for various combinations of drivers and then use univariate frequency analysis for the flood level.

- *Impact-Based Approach* (bottom-up approach): Extreme events from the response variable (e.g., flood level) are selected based on a threshold level (e.g., defined by the stakeholders). Identify coincident or nearly coincident values of the flood drivers to obtain a return period for the threshold.
- *Joint Density Approach*: Use a multivariate distribution and information on the dependence between the variables to extrapolate the input data to the extreme values. Multivariate parametric distributions and copulas are the most common tools. Multivariate parametric distributions are extensions of univariate distributions to higher dimensions. Because of several limitations, copulas are the preferred approach.

5.3 Multivariate Sampling

There are several approaches to sample data for a multivariate analysis:

- *Component-wise Block Maxima*: This method is the analogue of the univariate block maxima (e.g., the selection of maximum annual values). This technique will not necessarily capture simultaneously occurring bivariate events.
- *Threshold Exceedance*: Select a high threshold for each margin (u_x, u_y) and the model describes the dependence for extremes that simultaneously exceed both thresholds (see Fig. 5.2a). This technique cannot capture bivariate events with only one single extreme component.
- *Point Process*: Transform the data to radial $r = \sqrt{x^2 + y^2}$ and angular components and identify the extremes as events that occur above a high radial threshold r_0 (see Fig. 5.2b).
- *Conditional Extreme Events*: The distribution of Y is conditioned on the threshold exceedance in X and vice versa (see Fig. 5.2c).

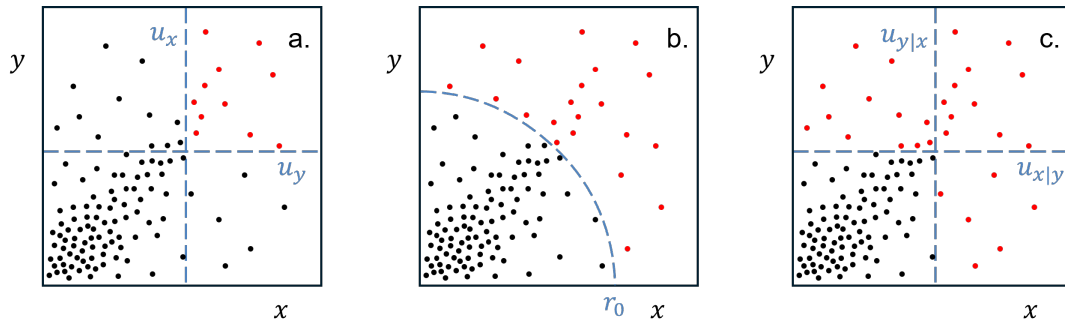


Figure 5.2. Visualization of the three multivariate sampling methods threshold exceedance (a.), point process (b.) and conditional extreme events (c.). The dashed blue lines represent the thresholds for the differentiation between extreme and not extreme events, with the dots highlighted in red representing the chosen data set of extremes.

5.4 Assessment of Dependence between Variables

There are three categories of dependence measures to assess whether there is a statistically significant correlation between the variables:

- *Linear Dependence*: A measure of the linear dependence between two variables can be assessed by computing the Pearson correlation coefficient ρ (seen in QQ-plots).

$$\rho = \frac{\text{Cov}(x, y)}{S_X \cdot S_Y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \cdot \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (5.5)$$

- *Rank (or Monotone) Correlation*: The non-linear dependence between two variables can be characterized by Spearman's ρ_s and Kendall's τ . This topic will be covered in more detail in module 3.

$$\rho_s = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 \cdot \sum_{i=1}^n (S_i - \bar{S})^2}} \quad (5.6)$$

where R_i and S_i represent the rank of x_i and y_i , respectively, and $\bar{R} = \bar{S} = (n + 1)/2$.

$$\tau = \frac{S}{n(n-1)/2} \quad \text{with} \quad S = \sum_{i < j} \text{sgn}(x_i - x_j) \cdot \text{sgn}(y_i - y_j) \quad (5.7)$$

- *Tail Dependence*: The conditional probability of two random variables taking extremes is measured by the tail dependence. These measures do not provide the strength of the association between the variables and are useful to summarize how extremes tend to occur simultaneously.

$$\chi_u(q) = \lim_{q \rightarrow 1} \mathbb{P} [y > G^{-1}(q) \mid x > F^{-1}(q)] \quad (5.8)$$

$$\chi_l(q) = \lim_{q \rightarrow 0} \mathbb{P} [y \leq G^{-1}(q) \mid x \leq F^{-1}(q)] \quad (5.9)$$

where $F(\cdot)$ and $G(\cdot)$ are the marginal distributions of x and y , respectively. For two independent variables, the test statistic is $\chi = 0$, whereas a $\chi = 1$ indicates a perfect (upper or lower) tail dependence.

5.5 Copulas

Copulas are tools that allow separate specification of the marginal distributions and the dependence structure of multivariate distributions. This allows the combination of different marginal distributions, which is generally one of the main limitations of multivariate distributions functions.

Sklar's theorem states that is F_{XY} is a joint distribution function with marginal distributions $u = F_X$ and $v = F_Y$, there exists a copula function C such that:

$$F_{XY}(x, y) = \mathbb{P}(X \leq x, Y \leq y) = C(F_X(x), F_Y(y)) \quad (5.10)$$

Sklar's inversion describes

$$C(u, v) = F_{XY}(F_X^{-1}(u), F_Y^{-1}(v)) \quad (5.11)$$

where F_X^{-1} and F_Y^{-1} represent the inverse of F_X and F_Y , respectively.

The copula function C let's us express the joint probability as a function of the marginal distributions. The value of the copula C as a function of u and v is visualized in Figure 5.3.

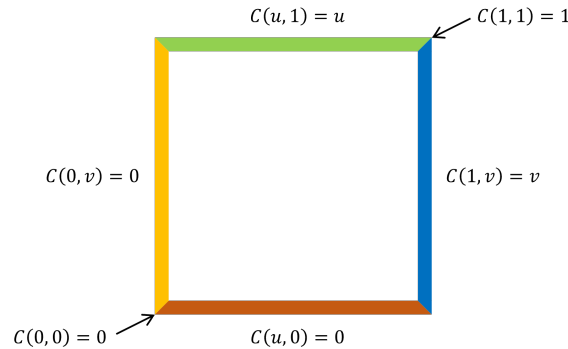


Figure 5.3. This figure illustrates the concept of a Copula, which separates marginal behavior from dependence in a bivariate distribution. The edges of the square represent the marginal distributions transformed to the uniform scale $[0,1]$, while the interior captures the dependence structure between variables.

5.5.1 Copula Families

There exist several different families of copulas to model different structures of dependence (e.g., absence or presence of symmetry, tail dependence) using varying levels of mathematical complexity.

- *Archimedean*: Widely employed copulas because of easy construction. This family includes Frank, Clayton, Gumbel, etc.
- *Elliptical*: Elliptically contoured
- *Empirical*
- *Vine (Pair)*: Graphical model

5.5.2 Parameter Estimation and Selection of Copulas

There are methods available to estimate the parameters of the copulas (Moment-like method, Maximum pseudo-likelihood method, Exact Maximum-likelihood methods). The selection is made comparing the maximized log-likelihood or Akaike Information Criterion *AIC* values.

$$AIC \text{ score} = 2k - 2\ln(L) \quad (5.12)$$

where k is the number of estimated parameters and L is the maximum value of the likelihood function.

Additionally, error metrics such as the root mean square error RMSE of the empirical and computed bivariate probabilities are employed to assist in the selection.

Bibliography

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A Appendix

A.1 Tables of Critical Values

Table A.1. Tabled values for the normal standard distribution, relating the curve left of the reduced variate z under the curve. For values with $z < 0$, employ the relation $\Phi(-z) = 1 - \Phi(z)$.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997

Table A.2. Table of the Student's t distribution for different degrees of freedom df and significance levels α .

df	α level									
	0.25	0.2	0.15	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.710	31.820	63.660	318.310	636.620
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
5	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
6	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
7	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
8	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
9	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
10	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
11	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
12	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
13	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
14	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
15	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
16	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
17	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
18	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
19	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
20	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
21	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
22	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
23	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
24	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
25	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
26	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
27	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
28	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
29	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
30	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
40	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
60	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
80	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
100	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
1000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291

A.2 Frequency Analysis - Standard Error

Table A.3. Mean μ_y and standard deviation σ_y of order statistics for various sample sizes n . Further details are explained in Kite (1977).

Sample Size	Mean	Standard Deviation
n	μ_y	σ_y
10	0.4952	0.9496
15	0.5128	1.0206
20	0.5236	1.0628
25	0.5309	1.0914
30	0.5362	1.1124
35	0.5403	1.1285
40	0.5436	1.1413
45	0.5463	1.1518
50	0.5485	1.1607
55	0.5504	1.1682
60	0.5521	1.1747
65	0.5535	1.1803
70	0.5548	1.1854
75	0.5559	1.1898
80	0.5569	1.1938
85	0.5578	1.1974
90	0.5586	1.2007
95	0.5593	1.2037
100	0.5600	1.2065

Table A.4. Parameter δ for the use in the computation of the standard error $S(x_T)$ of the Gumbel / EV-I distribution and Method of Moments for different return periods T and sample sizes n .

Sample Size n	Cumulative Probability $F_X(x)$ [-]					
	0.5	0.8	0.9	0.95	0.98	0.99
	Return Period T [a]					
	2	5	10	20	50	100
10	0.9305	1.8539	2.6199	3.3826	4.3869	5.1459
15	0.9269	1.7695	2.4756	3.1814	4.1127	4.8174
20	0.9250	1.7249	2.3990	3.0745	3.9670	4.6427
25	0.9238	1.6968	2.3506	3.0069	3.8747	4.5320
30	0.9229	1.6772	2.3169	2.9597	3.8103	4.4548
35	0.9223	1.6627	2.2919	2.9247	3.7624	4.3974
40	0.9218	1.6514	2.2725	2.8975	3.7252	4.3527
45	0.9214	1.6424	2.2569	2.8756	3.6954	4.3169
50	0.9211	1.6350	2.2441	2.8577	3.6708	4.2874
55	0.9208	1.6288	2.2333	2.8426	3.6502	4.2627
60	0.9206	1.6235	2.2241	2.8297	3.6326	4.2415
65	0.9204	1.6189	2.2162	2.8186	3.6173	4.2232
70	0.9202	1.6149	2.2093	2.8089	3.6040	4.2073
75	0.9201	1.6114	2.2032	2.8003	3.5923	4.1931
80	0.9199	1.6083	2.1977	2.7926	3.5818	4.1806
85	0.9198	1.6055	2.1928	2.7858	3.5724	4.1693
90	0.9197	1.6030	2.1884	2.7796	3.5639	4.1591
95	0.9196	1.6007	2.1844	2.7739	3.5562	4.1498
100	0.9195	1.5986	2.1808	2.7688	3.5491	4.1414

Table A.5. Parameter δ_m for the use in the computation of the standard error $S(x_T)$ of the Gumbel / EV-I distribution and the Method of Maximum Likelihood for different return periods T .

Cumulative Probability $F_X(x)$ [-]	0.5	0.8	0.9	0.95	0.98	0.99
Return Period T [a]	2	5	10	20	50	100
δ_m	0.7139	1.0956	1.4055	1.7195	2.1383	2.4574